Overview

You are tasked with designing, modelling, simulating, and verifying the behaviour of a Smart Intersection system using the UPPAAL model checker.

The intersection manages two perpendicular roads (North–South and East–West) and includes:

1. Traffic lights for vehicles.
2. Pedestrian lights for crossings.
3. Emergency vehicle detection and priority handling.

Your model must ensure both safety (no collisions, no unsafe crossings) and liveness (vehicles and pedestrians are eventually served) while handling the trade-off between pedestrian safety and emergency response time.

System Description

Intersection Layout

1. Two roads: North–South (NS) and East–West (EW).
2. One traffic lane in each direction for both roads.
3. Pedestrian crossings exist parallel to each road.
4. Each crossing has:
    a. One traffic light for vehicles.
    b. Pedestrian light for crossing pedestrians.

Traffic Light Operation

1. Traffic lights have three phases: Green → Yellow → Red.
2. NS and EW are never green simultaneously for vehicles.
3. The green phase lasts for a maximum of 1 minute in either direction.
4. A small all-red safety buffer exists before switching to another direction.

Pedestrian Crossing Operation

1. Pedestrians crossing parallel to traffic can walk in either direction during that traffic's green phase.
2. Pedestrian lights turn green only when perpendicular traffic is stopped.
3. The perpendicular road's traffic can only be stopped if the traffic has been moving for at least 15 seconds.
4. Crossing time is explicitly modelled:
    a. Each pedestrian takes 3 seconds to fully cross.
    b. Pedestrian lights must remain green for at least 15 seconds after a crossing starts.

Vehicle Movement

1. Vehicles pass through the intersection one at a time (one lane per direction).
2. Each vehicle takes 3 seconds to cross once the light is green.
3. Vehicles may arrive nondeterministically over time and queue up while the light is red.

Emergency Vehicle Priority

1. Emergency vehicles can arrive on either road at any time.
2. On detection, the system must preempt normal operation to allow immediate service to the emergency vehicle's road.
3. The corresponding light for this crossing needs to stay green until the emergency vehicle crosses the intersection.
4. Two possible preemption policies (you can choose and implement whichever you prefer):
    a. IMMEDIATE - Vehicle and pedestrian crossings may be interrupted to serve emergency vehicles as soon as possible.
    b. DEFERRED - Ongoing vehicle and pedestrian crossings always complete their minimum of 15s before switching to serve an emergency vehicle.
5. Safety constraints must be respected during preemption (no simultaneous greens, buffer delays before switching).

Modelling Requirements

Your UPPAAL model must include:

1. Separate timed automata for:
    a. Each traffic light (NS, EW).
    b. Each pedestrian crossing controller (NS, EW).
    c. Vehicle arrival and crossing logic.
    d. Pedestrian arrival and crossing logic.
    e. Emergency vehicle detector and preemption logic.
    f. Central controller (or synchronisation scheme) coordinating light states.
2. Explicit timing constraints for:
    a. Green, yellow, and all-red phases.
    b. Pedestrian crossing minimum green time.
    c. Vehicle and pedestrian crossing durations.
3. Queues for vehicles and pedestrians (at least as integer counters).
4. Channel synchronisation between automata for:
    a. Light changes.
    b. Emergency preemption.
    c. Queue updates.

Simulation Requirements

You must simulate:

1. Normal traffic operation with random arrivals of vehicles and pedestrians.

2. Emergency vehicle arrival during:
    a. Vehicle green phase in the opposite direction.
    b. Ongoing pedestrian crossing.
3. Queue build-up under heavy traffic conditions.
4. Effect of your chosen preemption policy (IMMEDIATE or DEFERRED) on:
    a. Emergency vehicle wait time.
    b. Pedestrian interruption rate.
    c. Vehicle throughput.

Verification Requirements

Using UPPAAL's model checker, verify the following

Safety properties

1. The traffic light for NS and EW is never green simultaneously.
2. The pedestrian light for one direction and the traffic light for its perpendicular direction are never green simultaneously.

Liveness properties

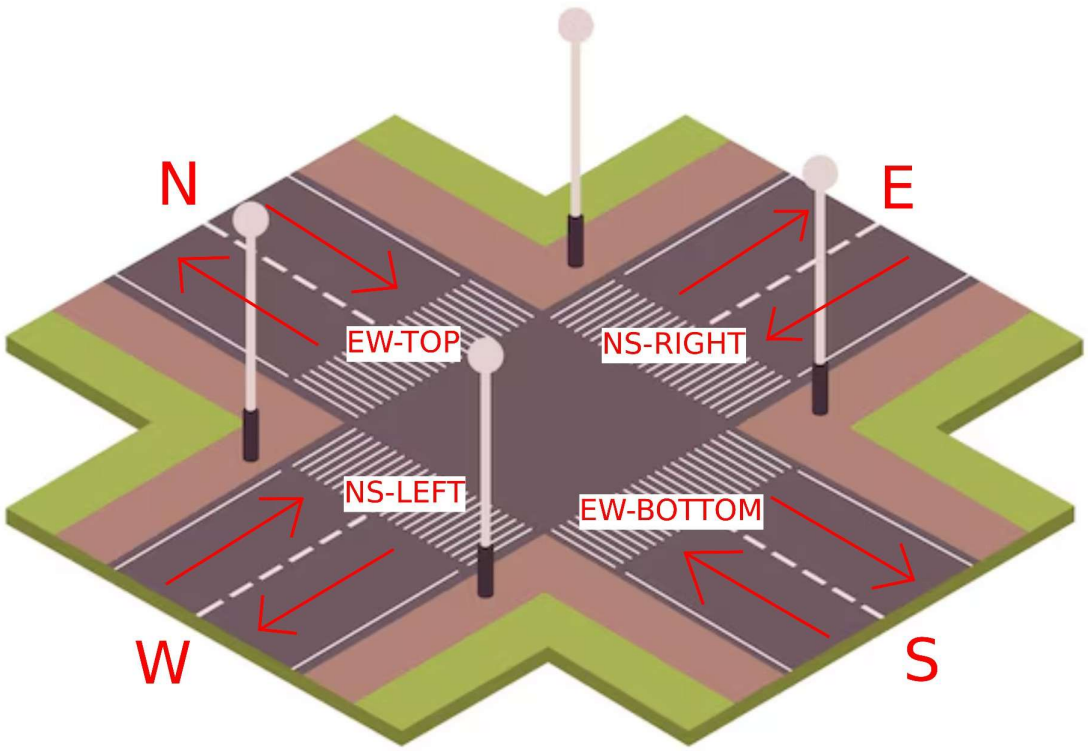
1. All vehicles are eventually allowed to cross.
2. All pedestrians are eventually allowed to cross.

Policy-dependant properties

1. IMMEDIATE:
    a. There exists a sequence of events where vehicle and pedestrian greens are interrupted by emergency preemption.
    b. Emergency vehicles receive green within 5s (since any vehicle/pedestrian crossing the road needs to finish crossing before the light for the emergency vehicle turns green).
2. DEFERRED:
    a. Pedestrian and vehicle greens are never shorter than 15s.
    b. Emergency vehicles receive green within 20s (15s + above reasoning).

Simulated Examples

Refer to the below image for a view of the intersection when it is empty



Regular arrival of cars without any pedestrians

1. t=0s
   NS light is green, EW light is red, NS-LEFT and NS-RIGHT lights are also green. There are no cars on the road.
2. t=1s
   One car arrives at the bottom of the intersection.
3. t=2s
   Another car arrives at the bottom of the intersection.
4. t=4s
   The first car finishes crossing the intersection.
5. t=5s
   The second car finishes crossing the intersection. Imagine that an infinite number of cars queue up at both W and E.

6. t=60s
   The NS light turns yellow. Since there are no cars in transit, it immediately turns red. EW light turns green.
7. t=63s
   The first car from the W and E queues crosses the intersection.
8. t=120s
   EW light turns yellow. No more cars are allowed into the intersection, but there are cars in transit. The car that left just before the EW light turned yellow will take 3 more seconds to cross the intersection, so we have to wait for it to cross.
9. t=123s
   EW light turns red. The NS light turns green.

Regular arrival of cars with pedestrians

1. The scenario is the same as the first till t=63s.
2. t=64s
   A pedestrian arrives that wants to cross using the zebra crossing labelled “NS-LEFT”. The pedestrian requests to cross.
3. t=75s
   Since a pedestrian requested to cross, the EW light turns yellow just 15s after it turned green. Following the same logic as before, we must wait for 3 more seconds before the light turns red.
4. t=78s
   EW light turns red. The NS-LEFT, NS-RIGHT and NS lights turn green. NS-RIGHT and NS are turned green since the perpendicular traffic that was obstructing them has been stopped. The pedestrian can now cross.
5. t=80s
   A pedestrian arrives that wants to cross using the zebra crossing labelled “EW-BOTTOM”. The pedestrian requests to cross.
6. t=93s
   The NS light turns yellow. Since there are no cars in transit, it immediately turns red. The EW-BOTTOM, EW-TOP and EW lights turn green. The pedestrian can now cross.

Regular arrival of cars with pedestrians and emergency vehicles (with IMMEDIATE policy)

1. The scenario is the same as the previous one till t=93s.
2. t=100s
   An emergency vehicle arrives at S, trying to cross to N. Immediately, the EW, EW-TOP and EW-BOTTOM lights all turn yellow.
3. t=103s
   The above lights turn red. The NS light turns green.

Regular arrival of cars with pedestrians and emergency vehicles (with DEFERRED policy)

1. The scenario is the same as the previous one till t=93s.
2. t=100s
   An emergency vehicle arrives at S, trying to cross to N.
3. t=108s
   The EW, EW-TOP and EW-BOTTOM lights turn yellow.
4. t=111s
   The above lights turn red. The NS light turns green.

Additional Resources

1. https://homes.cs.aau.dk/~bnielsen/MTV08/material/uppaal-tutorial.pdf
2. https://docs.uppaal.org/gui-reference/concrete-simulator/
3. https://docs.uppaal.org/language-reference/expressions/